A Falsifiable, Multi-Paradigm Pipeline for Bioacoustic Structure Discovery

Sheel Morjaria

Abstract

Discovering structure in animal vocalizations without anthropomorphic contamination remains a

1. Introduction

Animal vocalizations carry rich acoustic structure. Songs, calls, and bouts exhibit recurrenc

We address these challenges with a pipeline built on three design principles:

1. **Schema-first falsifiability.** Every output is a six-axis Evidence Profile validated by a sch

2. **Ethological firewall.** Every model in the pipeline carries machine-readable provena

3. **Semantic ceiling.** Acoustic tokens (VQ-VAE or k-means clusters) are acoustic

Contributions

This work makes the following contributions:

A **segmentation router** that selects the appropriate segmenter per acoustic regime (PCEN+C

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2. Related Work

Spoken-term discovery (Park & Glass, 2008; Jansen et al., 2013) finds recurring acoustic p

Bioacoustic structure analysis has typically employed either hand-crafted features

(spectral,

Network analysis of vocal sequences (Sasahara et al., 2012; Kershenbaum et al., 2014)

models ar

Recurrence quantification analysis (Webber & Zbilut, 1994; Marwan et al., 2007)

quantifies

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3. Framework

3.1 The Segmentation Router

Different acoustic regimes require fundamentally different segmenters. A single-segmenter

approach

The router maintains three regimes in `configs/segmentation_router.yaml`:

Regime	Method	Strength	Example
Discrete calls in noise	PCEN+CRNN → HECB annotation	Isolates calls from noise floor	MeerKAT (8 kHz)
Continuous speech-like	NBD + CPC calibrator (ABSTAIN)	Precision via learned context	LibriSpeech (16 kHz)
Exploratory / unknown	Fixed-threshold NBD + ABSTAIN	Conservative; no false structure	Novel species

The router decision is auditable: every run produces `trace["router_audit"]` recording the

detected

3.2 HECB-Seg: Hierarchical Event-Conditioned Boundary Segmentation

For calls-in-noise, the pipeline uses a two-stage architecture:

Stage 1 — Call isolation: PCEN spectrogram → Conv2D→BiLSTM → P(call) per frame →

hysteresis

Stage 2 — Internal annotation: NBD proposals inside each call event → typed classifier

(onset/off

A parent-quality gate (`min_parent_confidence`) abstains on low-confidence parent events,

preventin

3.3 DUSTED: DTW Recurrent Acoustic Terms

DUSTED adapts spoken-term discovery to animal vocalizations. Each bout or call is represent

Validated on three structural regimes (uniform DTW threshold thr=0.15, HAC clustering):

Corpus	Regime	# Families	Coverage	Null ratio	Bootstrap
Bengalese finch	Discrete, diverse	457	0.480	**2.67×**	0.509
Egyptian fruit bat	Graded, continuous	306	0.153	1.36×	0.394
Infant marmoset	Sparse, degenerate	139	0.028	1.55×	0.229

3.4 The Acoustic Fingerprint (44-D, All-Species)

The pipeline's cross-species representation is a physics-grounded, species-portable per-segm

Physics (~16D): spectral centroid, bandwidth, rolloff, flatness, flux, zero-crossing rate, RM

Voiced/harmonic features are flagged and zeroed for non-tonal segments (anti-anthropomorphism).

Mel-MFCC

3.5 Ethological Firewall

Every model in the pipeline carries provenance metadata in `configs/model_metadata.json`: pretrainin

1. Text-supervised models (NatureLM) are retrieval-only.

2. Cross-s

3.6 The Six-Axis Evidence Profile

The verdict is six **independent** axes, each categorical with a calibrated confidence:

`sequence_recombination · functional_aliasing · call_unitization · context_binding · turn_struc

Scorers take primitive signals (LZ complexity, alias density, Cramér's V, etc.) so the profile is u

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4. Results

4.1 What Acoustics Encode: Individual Identity and Call Form

Corpus	Target	Metric	Result	Lift
Egyptian fruit bat	Emitter identity	Cramér's V	0.446	—

Rook	Source (individual) identity	Cramér's V	0.31	—
Marmoset	Call type	Cramér's V / linear probe	0.599 / 0.829	3.0× / 4.1×
MeerKAT	Call type	Cramér's V / linear probe	0.521 / 0.753	—
Bengalese finch	Syllable (GT)	DTW purity (wmean)	0.788	—

Acoustic structure reliably carries **individual identity** (bat, rook) and **call/event form** (m

4.2 What Acoustics Do Not Encode: Context, Valence, and Meaning

Finding	Evidence	Section
Context requires social covariates	Bat: D(acoustic+social) = 4.3× ≫ A(acous	§9
Valence is locked in call type	Soundwel: within-call-type fp→valence 0.	§24
Social partner ≈ chance	RookID: cross-caller retrieval below cha	§26
Tokens ≠ words	LibriSpeech: token↔word macro consistenc	§29
Degenerate repertoire (token-free)	Marmoset MP: 87.5% stutter-dominant, us	§34

The **semantic ceiling** holds across all species: acoustics alone cannot recover context, emotional

4.3 Multi-Paradigm Structural Corroboration

To demonstrate that DUSTED's findings are not method-dependent, four independent mathema

Paradigm	Mathematics	Dataset	Key Result	Verdict
PST + Transition Network	Variable-order Markov + graph topology	Bengalese finch (discrete)	247 predictive contexts, 5 communities	**Corroborates** DUSTED's discrete-
RQA	Recurrence plot → %DET, %LAM	Bat (GM)	%LAM context-dependent (Kruskal-Wallis p	**Complements** DUSTED where DT
Matrix Profile	STUMP continuous subsequence search	Marmoset (sparse)	87.5% stutter-dominant, 0 motifs	**Confirms** degenerate repertoire t
Transition Network	Networkx community detection	LibriSpeech (diverse)	639 word nodes, 19 communities,	**Universality** under scales to human-s

Three paradigms (alignment, graph, dynamics) agree on the biological reality of the three repertoire

4.4 Negative Controls: Pipeline Specificity

To counter the reviewer argument that "the pipeline finds structure everywhere," three negative

Control	Description	# Motifs	vs Natural (33)
Synthetic FM chirps	Non-biological FM-sweep sequences	1	33× specificity
Random audio segments	Random chunks from bat recordings	1	33× specificity
Order-shuffled bat	Same units, permuted order	22	Partial collapse

The pipeline finds **1 motif** in synthetic and random audio versus **33** in natural bat bouts—a

4.5 Boundary-Quality Sweep: Whole-Call Primary

The boundary-quality sensitivity sweep varies parent-event boundary quality (oracle GT → CRNN at

4.6 The Entanglement Ceiling: A Mathematical Proof of the Semantic Ceiling

§4.1 showed that both caller identity and call form are reliably encoded in the acoustic

fingerprin

We attempt to scrub caller identity from the 44-D acoustic fingerprint by four methods

spanning

Method	Mathematics	Δ caller V	Δ form V	Result
LDA projection	Linear subspace	−0.029	**−0.120**	FAIL (subspaces overlap)
LogReg projection	Linear weight	−0.005	−0.013	FAIL (inseparable)
Z-normalization	Affine (mean + scale)	−0.019	**−0.074**	FAIL (overlap)
Delta-trajectory	Non-linear differencing	−0.105	**−0.163**	FAIL (dynamics are caller-specific)

Biological interpretation: the medium is the message. The delta-trajectory failure is

the most

Human vs. animal. This closes the loop on the cross-species analogy. Human speech *can*

be partial

Architectural consequence. This elevates the pipeline's invariants from good engineering

practice t

The firewall is proven, not policy. Because linear algebra cannot disentangle animal

identity

Learned-representation control (AVES, §36). We repeated the linear disentanglement on

AVES, a 7

Consequently, bioacoustic semantic decoding from acoustics alone is not merely difficult — it is algeb

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5. The H0-CMC Evidence Ladder

The Causal Multimodal Convergence (H0-CMC) framework defines a six-rung evidence ladder connectin

Tier	Evidence	Autonomous?	Semantic validation
T0 — Acoustic	Recurrent acoustic structure found	✓	none
T1 — Context	Context-associated cue (binding signific	✓	none (correlation)
T2 — Playback	Counterfactual stimuli generated, respon	✓ (response pending)	none (response pending)
T3 — Convergence	Multimodal convergence (acoustic + behav	✓	candidate_functional_interpretation
T4 — Counterfactual	Counterfactual disrupts receiver respons	✗ (requires live	causally_supported_signal
T5 — Field	Ecologically validated (fitness-relevant	✗	ecologically_validated_function

The pipeline operates autonomously at T0–T1. It generates T2 playback stimuli via the DSP Audio

An explicit-implicit test classifies each corpus's communication mode: if acoustics alone recover c

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6. Track C: Playback Readiness

The DSP Audio Splicer (`experiments/playback_design/audio_splicer.py`) generates falsifiable

Condition	Transform	Hypothesis	H0-CMC tier
Natural	Unchanged baseline	—	—
Order shuffled	Permute unit order	Does sequence syntax matter?	T2
Onset–offset swap	Swap first and last units	Do positional roles matter?	T2
Caller swapped	Replace one unit with a donor from another	Does caller identity matter?	T2
Amplitude matched	RMS-normalize to target	Control: rules out loudness confound	—

Stimuli are generated via exemplar-based DSP (slice + crossfade + concatenate), not neural synthesis

A compositional extension (`generate_compositional_counterfactuals`; §37) tests whether a sequence

7. Discussion

7.1 The Three-Bucket Thesis

Every finding in this pipeline lands in exactly one of three buckets:

1. **What acoustics encode.** Individual identity (bat Emitter V = 0.446, rook Source V = 0.31) ar

2. **What acoustics do not encode.** Context (bat requires social covariates; social_lift = 1.59),

3. **What requires social or biological validation.** All functional interpretation. The H0-CMC

7.2 Honest Limitations

Tokenization. DUSTED operates on k-means cluster IDs from the acoustic fingerprint, not learn

7.3 What This Pipeline Is Not

This pipeline is explicitly **not a classifier** that emits labels. It does not claim validated

8. Conclusion

We have presented a falsifiable, multi-paradigm pipeline for bioacoustic structure discovery

Reproducibility

All code, tests (247), and validated results (35 benchmark sections) are in the `bioacous

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